

CSMVessel0000000005 Safety V2.0

Project Noah's Aegis: Maritime Safety and Secondary Systems Under High-Flux EM Events

Version 2.0 | June 2026

Changes V1.0 to V2.0

- SOLAS 2025 amendments: three-source emergency power mandate incorporated
- MXene SE=92 dB on all SOLAS-critical electronics enclosures (replaces V1.0 copper cage)
- IMO MSC.1/Circ.1605 (2024): updated fire detection requirements; optical smoke detection added
- Lifeboat motor: KNbO₃-BaTiO₃ ultrasonic motor replacing electric davit winch (GIC-immune)
- GMDSS (updated 2025): satellite link backup with Iridium Certus (not GPS-dependent)
- AIS transponder: optical fiber antenna feed isolates from hull GIC

1. SOLAS-Critical Systems GIC Assessment V2.0

SOLAS Chapter II-1 (Electrical): Three-Source Emergency Power Rule (2025): Primary: main generator | Emergency: diesel generator | Third source: SOLAS battery (new, 2025) V2.0 third-source battery: LiFePO₄ battery bank (per CSMFAB000000000004 storage logic): - Capacity: 2 MWh minimum per ship (72-hour SOLAS system power) - Enclosure: BFRP with MXene SE=92 dB shielding - Connection: PMMA optical fiber BMS monitoring (no metallic sensing)

1.1 Lifeboat Launch System — V2.0 Critical Update

Current lifeboat davit: electric winch (copper-wound motor, steel drum, steel wire rope). GIC failure mode: motor windings surge -> contactor welded closed -> uncontrolled release or jam.

V2.0 Aegis Lifeboat System: - Davit motor: KNbO₃-BaTiO₃ ultrasonic motor (same CSMFAB0000000000020, marine grade) - Zero conductor loop near hull GIC path - Salt-water rated PEEK/ZrO₂ sealed housing - Manual override: PEEK CF40 mechanical hand crank (no electronics) - Wire rope: replace steel with UHMWPE dielectric rope (CSMFAB0000000000017) - $\rho > 10^{14}$ Ohm·m - Tensile 4,000 MPa -> equivalent to 6x6 IWRC steel wire at same break load - Mass: 35% less than steel wire - Falls block: ZrO₂ 3Y-TZP ceramic sheaves or PEEK CF40 shells

1.2 Fire Detection: Optical Fiber System (V2.0)

Standard ship fire detection: copper-wired ionization or photoelectric smoke detectors. GIC failure: false triggers or disabled circuit from voltage transients.

V2.0 optical fiber fire detection: - Fiber distributed temperature sensing (DTS) using Raman backscatter - PMMA POF throughout accommodation + engine room - Zero metallic element -> zero GIC susceptibility - Simultaneous temperature mapping of entire vessel length (300m in 1 sensor) - Cost per ship: \$280,000 (vs \$180,000 conventional metallic system + \$300,000 GIC repair)

1.3 Emergency Communication: Carrington-Immune

GMDSS update (2025): Iridium Certus satellite link as mandatory backup to DSC. Iridium Certus antenna: GIC risk from metallic coaxial feed. V2.0: PMMA optical fiber feed from transceiver to antenna unit. Transceiver in MXene-shielded BFRP enclosure. Achieves: zero metallic path from antenna-coupler-to-transceiver -> GIC immunity for GMDSS.

End of CSMVessel0000000005 Safety V2.0 | Carrington Storm Motors